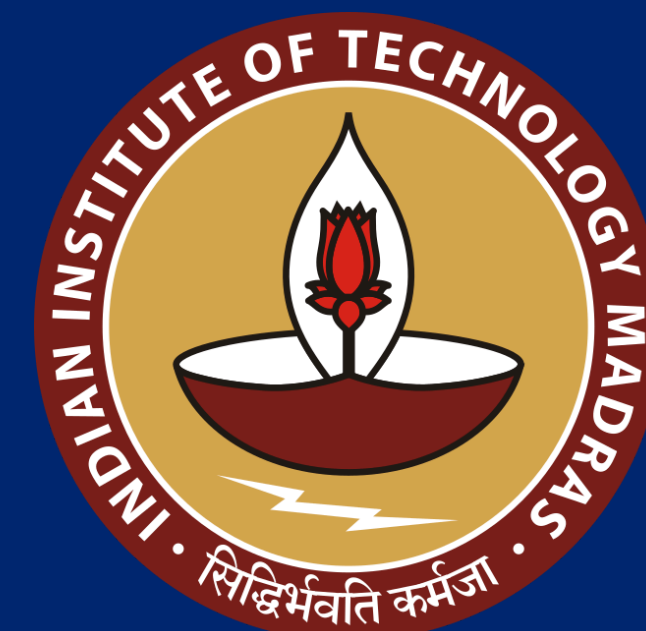




Modelling the acoustic response of a model combustor

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1. Abstract

Many central issues that arise in the development of modern combustion systems are due to the coupling mechanisms between the flames and the acoustic waves. These range from environmental issues such as noise to destructive mechanisms that can be generated in combustors. A resonant interaction may occur between the flame and the acoustic waves, which subsequently leads to combustion instabilities in many systems. Considerable research has gone into studying the effects of combustion instabilities and noise along with methodologies to mitigate their effects [1,2]. This work focuses on studying the acoustics in a simplified combustor model using a commercial Helmholtz solver (COMSOL) to determine the frequencies along with the mode shapes of the resonant instabilities. It is assumed that there is no mean flow in the combustor model. The media in the system is taken to be entirely air as a reference case, following which the media is taken to be reactant and product gas mixtures in the appropriate regions of the computational domain. In the early stages of combustion noise research, open turbulent flames were identified primarily as acoustic radiators exhibiting monopole-like behavior. The resulting acoustic field was generally attributed to a distribution of monopole sources embedded throughout the flame region [3]. A Gaussian distribution over a spherical domain was assumed for the source term, which was used to represent the flame in the combustor. The Green's function in frequency space was constructed for the system which represents the effects of the acoustic radiation from an individual source at any other location in the domain. The solution for the pressure fluctuations in the frequency space was obtained using the Green's function constructed for the geometry of the model combustor and was compared against the results obtained from the Helmholtz solver, thereby enabling the development of a reduced-order model based on the Green's function approach [4]. The results were compared for different types of fuels of the same equivalence ratio to determine the influence of fuel composition on the acoustic response in the model combustor.

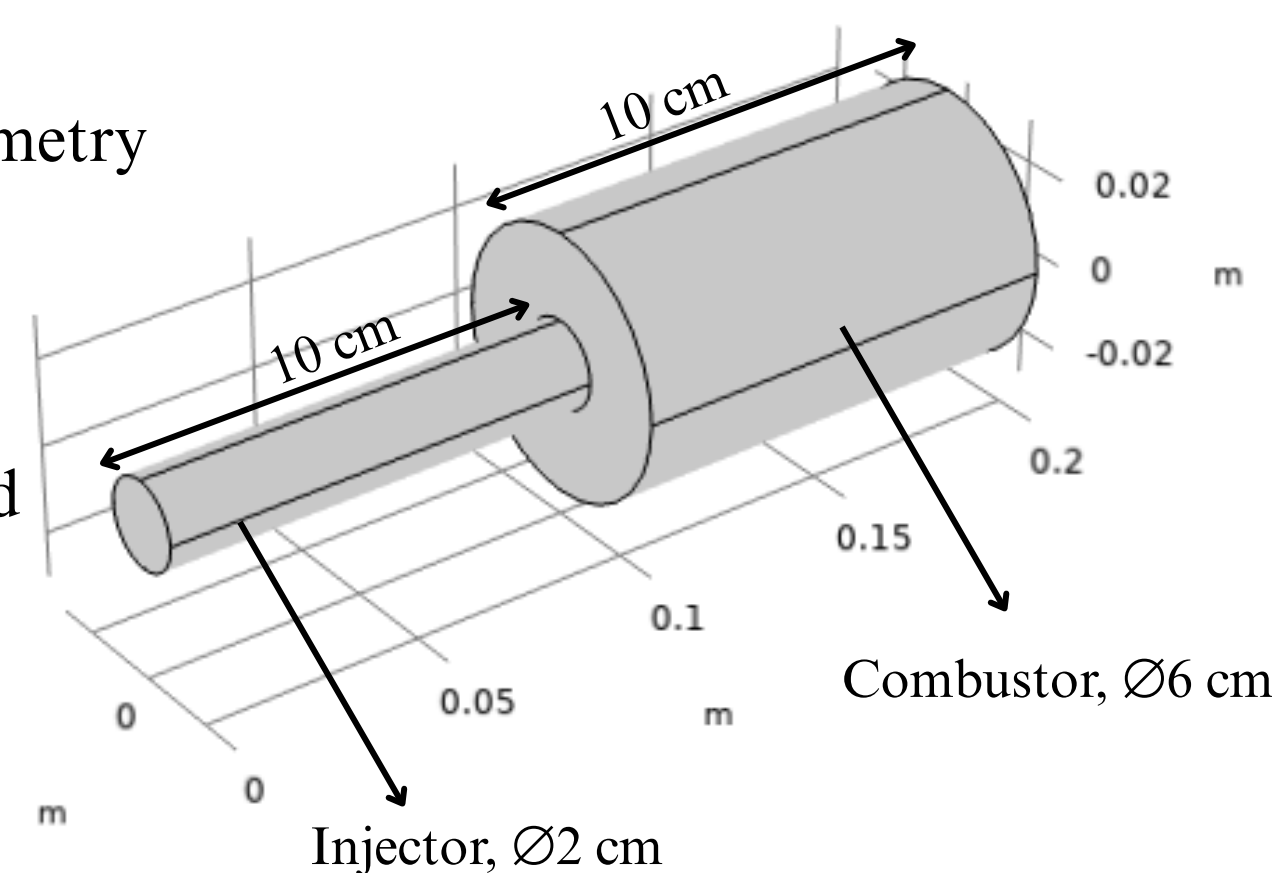
2. Computational Domain and Method

• Computational Domain:

- Combustor model: double duct geometry
- Smaller duct: injector
- Larger duct: combustor

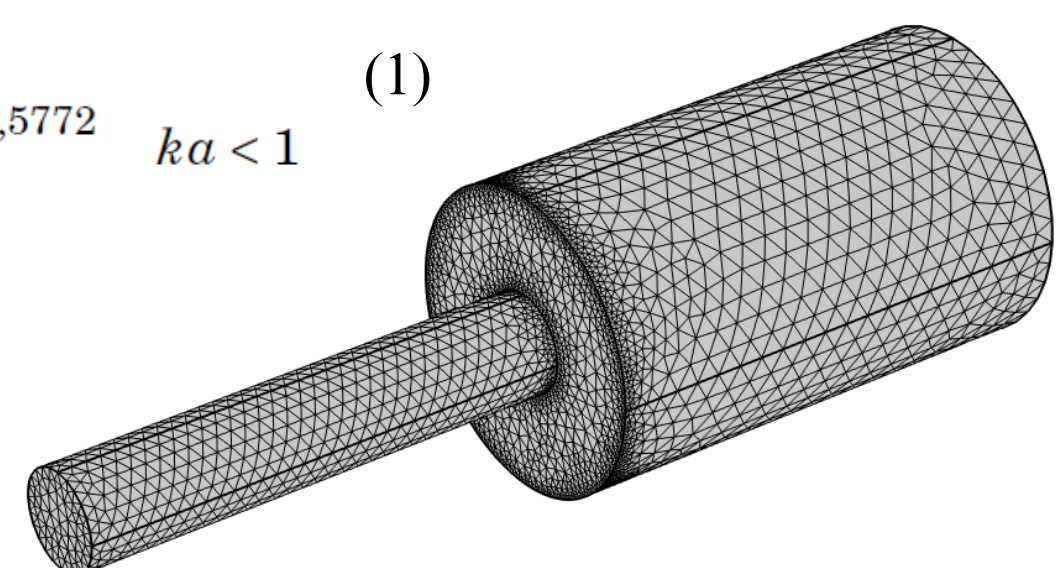
• Boundary Condition:

- Curved surfaces, junction: sound hard walls.
- Inlet, Outlet: Radiation impedance conditions as given below [5]



$$Z_{\text{end}} = \rho c \frac{1+R}{1-R} \quad R = |R|e^{2ika\delta} \quad (1)$$

$$|R| = e^{-(ka)^2/2} \left(1 + \frac{1}{6}(ka)^4 \left[\ln\left(\frac{1}{\gamma ka}\right) + \frac{19}{12} \right] \right) \quad \gamma = e^{0.5772} \quad ka < 1$$



• Meshing:

- Free tetrahedral mesh elements
- Finer mesh near the sudden expansion

• Validation:

Theoretically eigenmodes can be determined by the **Transfer matrix method** [1].

$$A_1^+ / A_1^- = R_1 \quad \begin{bmatrix} A_2^+ \\ A_2^- \end{bmatrix} = T \begin{bmatrix} A_1^+ \\ A_1^- \end{bmatrix} \quad T = \begin{bmatrix} e^{ika_1}(1+\Gamma) & e^{-ika_1}(1-\Gamma) \\ e^{ika_1}(1-\Gamma) & e^{-ika_1}(1+\Gamma) \end{bmatrix}$$

$$A_2^+ / A_2^- \cdot e^{(2ika_2)} = R_2 \quad \Gamma = \frac{S_1}{S_2} \quad (2)$$

Mode	Theoretical Eigenfrequency (Hz)	Eigenfrequency from Simulation (Hz)
0	148.63+15.48i	161.69 + 16.36i
1	821.19-0.79i	892.63 + 4.5546i
2	1863.63+15.48i	1876.4 + 16.282i
3	2536.19-0.79i	2535.7 + 4.6106i
4	3578.63+15.48i	3591.1 + 16.18i

• Flame Model:

The flame is modelled as a Gaussian distribution of monopole sources over a spherical region of radius 3 cm inside the combustor given below [3].

$$\nabla \cdot \left(-\frac{1}{\rho_c} (\nabla p_t - \mathbf{q}_d) \right) - \frac{k_{eq}^2 p_t}{\rho_c} = Q_m \quad Q_m = i\omega Q_{\text{heat}} \frac{\alpha_p}{\rho C_p}, \quad \alpha_p = \frac{1}{c} \sqrt{\frac{C_p(\gamma-1)}{T}}$$

$$Q_{\text{heat}} = 0.1 \exp \left(-\log(2) \cdot \frac{(x-0.115)^2 + y^2 + z^2}{\sigma^2} \right) \quad (3)$$

• Green's Function Formulation:

Constructing the green's function ($N_{\text{max}} - 8$):

$$\hat{G}(\mathbf{x}, \mathbf{x}_0, \omega) = \sum_{n=1}^{N_{\text{max}}} \frac{\Phi_n(\mathbf{x}) \Phi_n(\mathbf{x}_0)}{V \Lambda_n (k_n^2 - k^2)} \quad \Lambda_n = \frac{1}{V} \int_V \Phi_n(\mathbf{x}_0) \Phi_n(\mathbf{x}_0) d\mathbf{x}_0 \quad (4)$$

Determining pressure fluctuations:

$$\hat{p}(\mathbf{x}, \omega) = i\omega \int_V \frac{\gamma}{c^2} \hat{G}(\mathbf{x}, \mathbf{x}_0, \omega) \hat{Q}(\mathbf{x}_0, \omega) d\mathbf{x}_0 \quad (5)$$

• Fuel Compositions:

- Fuels considered - Hydrogen, ammonia, 1:1(by moles) mixture of hydrogen and ammonia
- Equivalence ratio used - **0.6**, eigenfrequency analysis does not show much variation for change in equivalence ratio

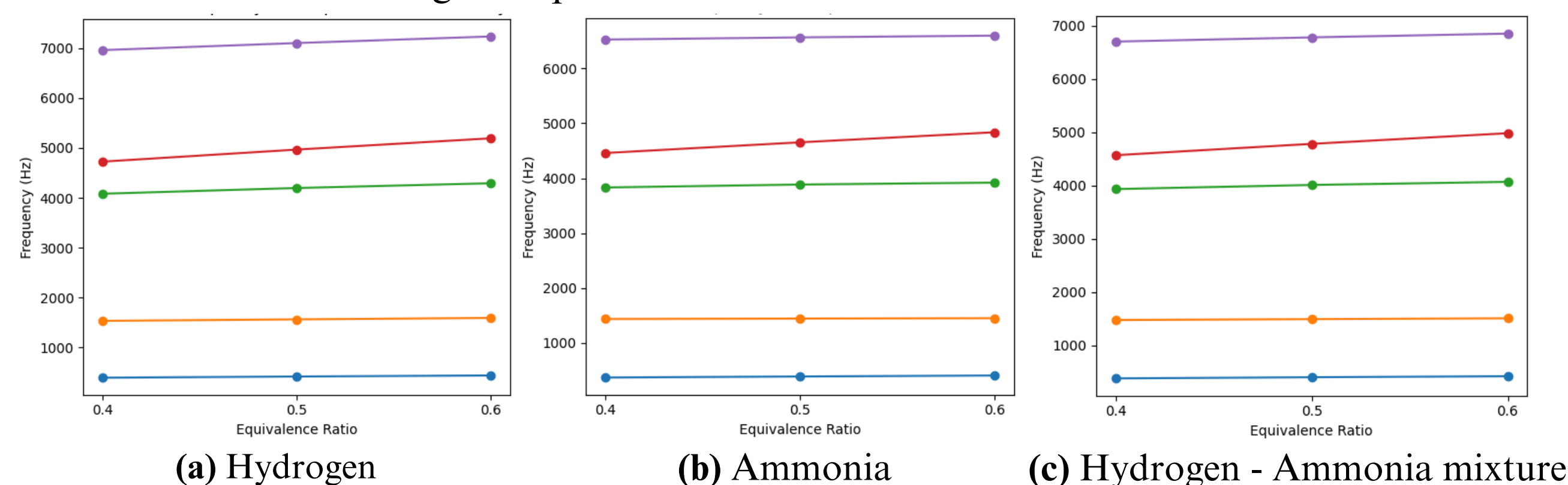


Figure 1: Variation of eigenfrequency with equivalence ratio for different fuels

List of Symbols

Z	Impedance	γ	Specific heat ratio
R	Reflection Coefficient	c	Speed of sound
p_t	Total Pressure	α_p	Thermal expansion coefficient
Q_m	Monopole Source	C_p	Heat capacity at constant pressure
Φ_n	n^{th} Eigenfunction		

3. Results

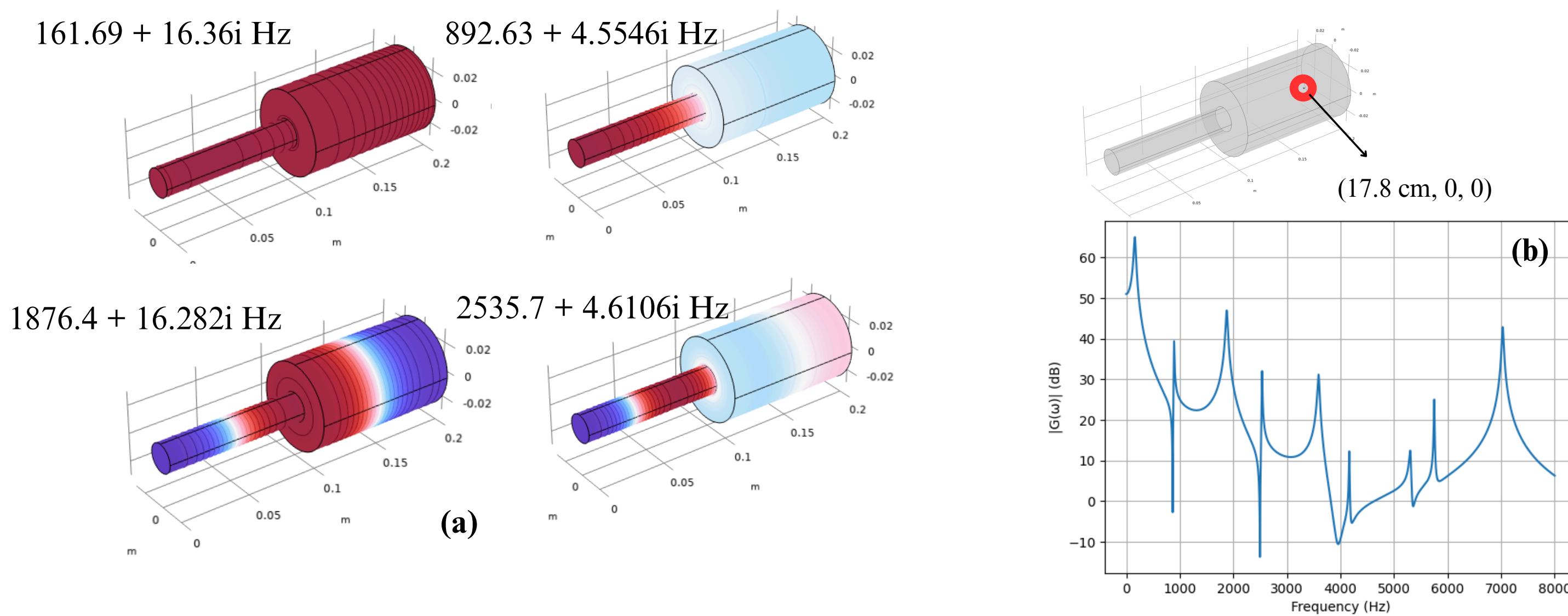


Figure 2: (a) First four eigenfunctions with (b) corresponding Green's function between the flame and the location ($x = 17.8$ cm, $y = 0$, $z = 0$) as indicated in the figure (Medium: air)

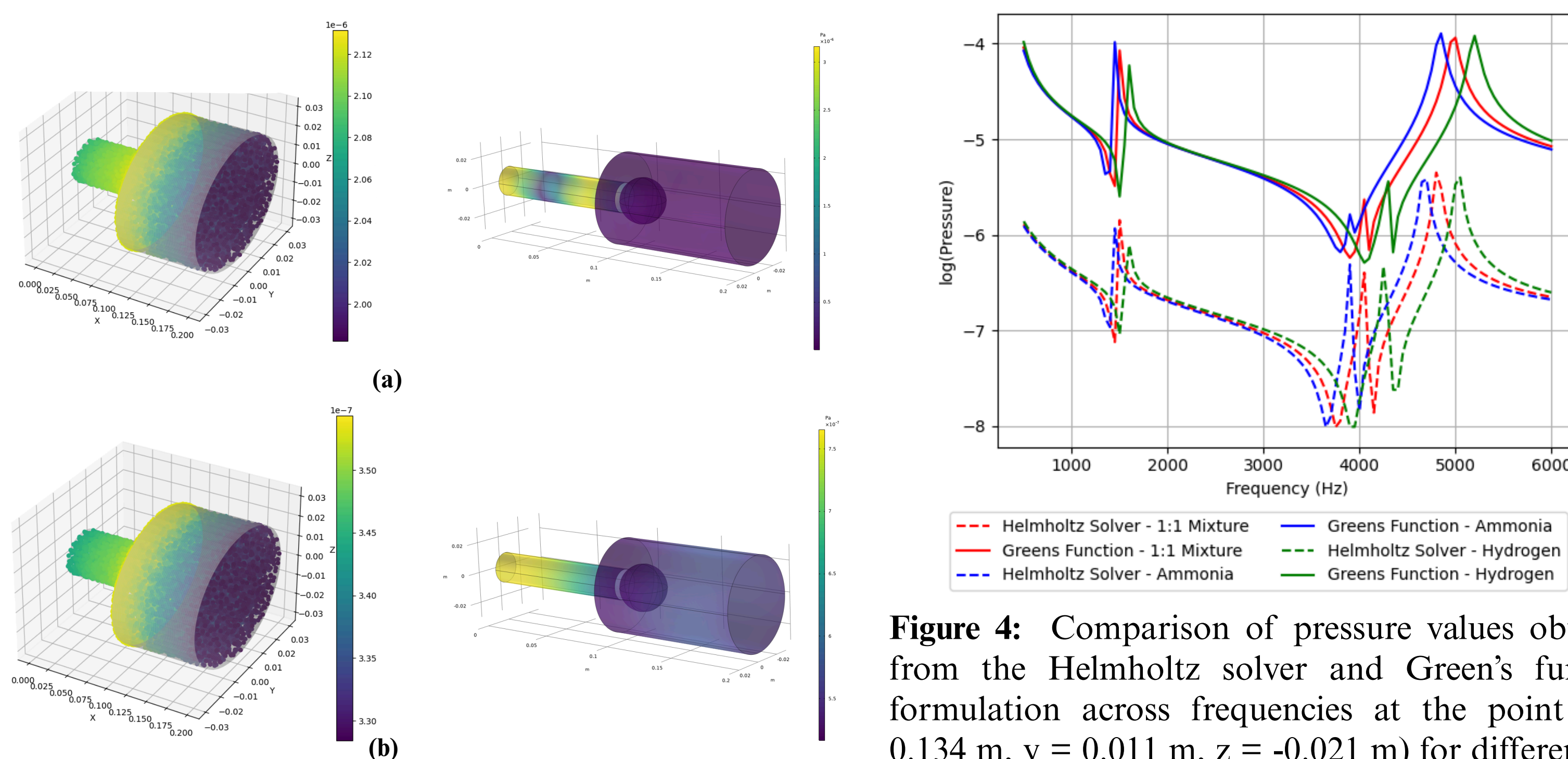


Figure 3: Comparison of pressure field from the Helmholtz solver (right) and Green's function formulation (left) for (a) 2500 Hz and (b) 500 Hz (Medium: air)

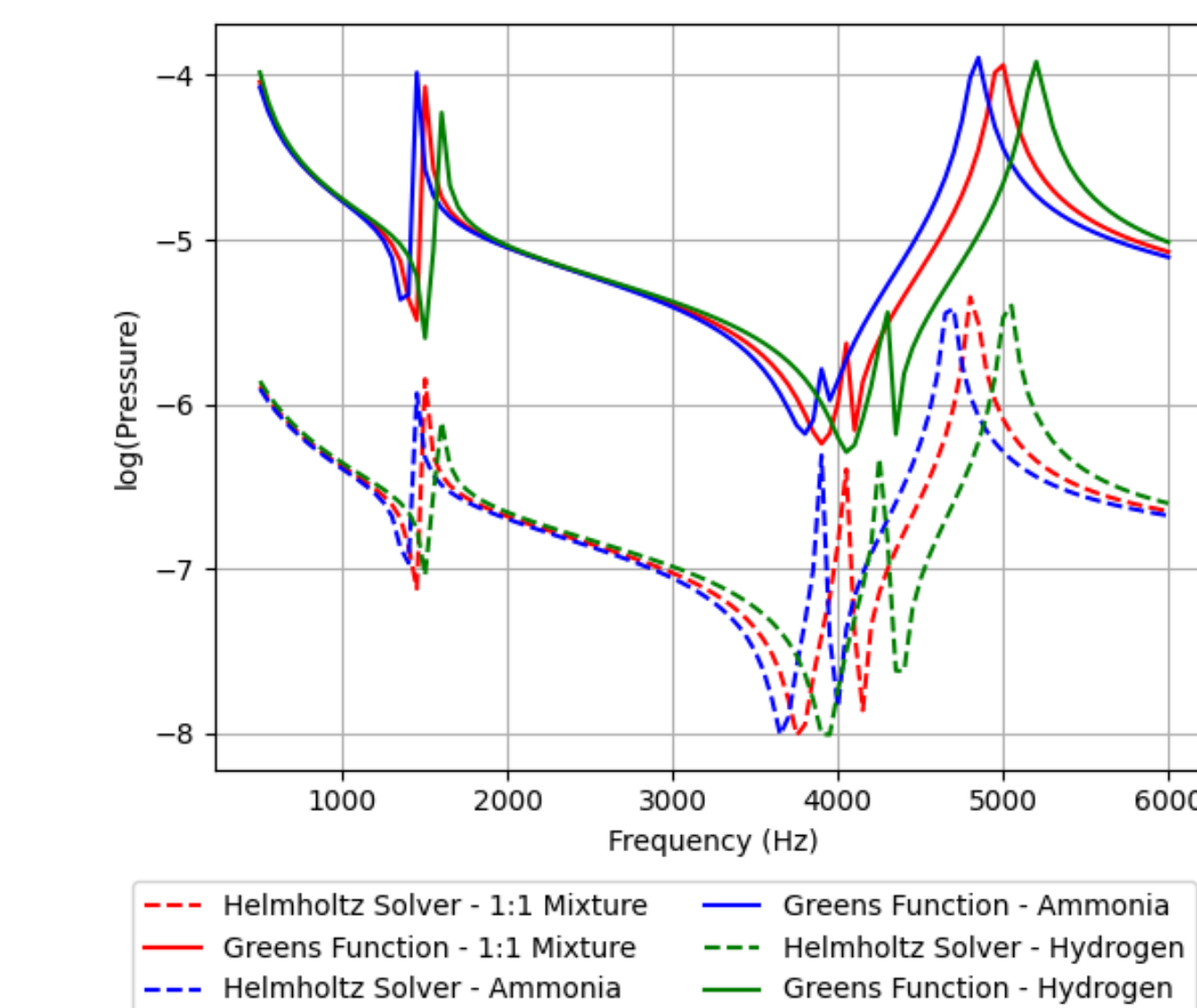


Figure 4: Comparison of pressure values obtained from the Helmholtz solver and Green's function formulation across frequencies at the point ($x = 0.134$ m, $y = 0.011$ m, $z = -0.021$ m) for different fuel compositions

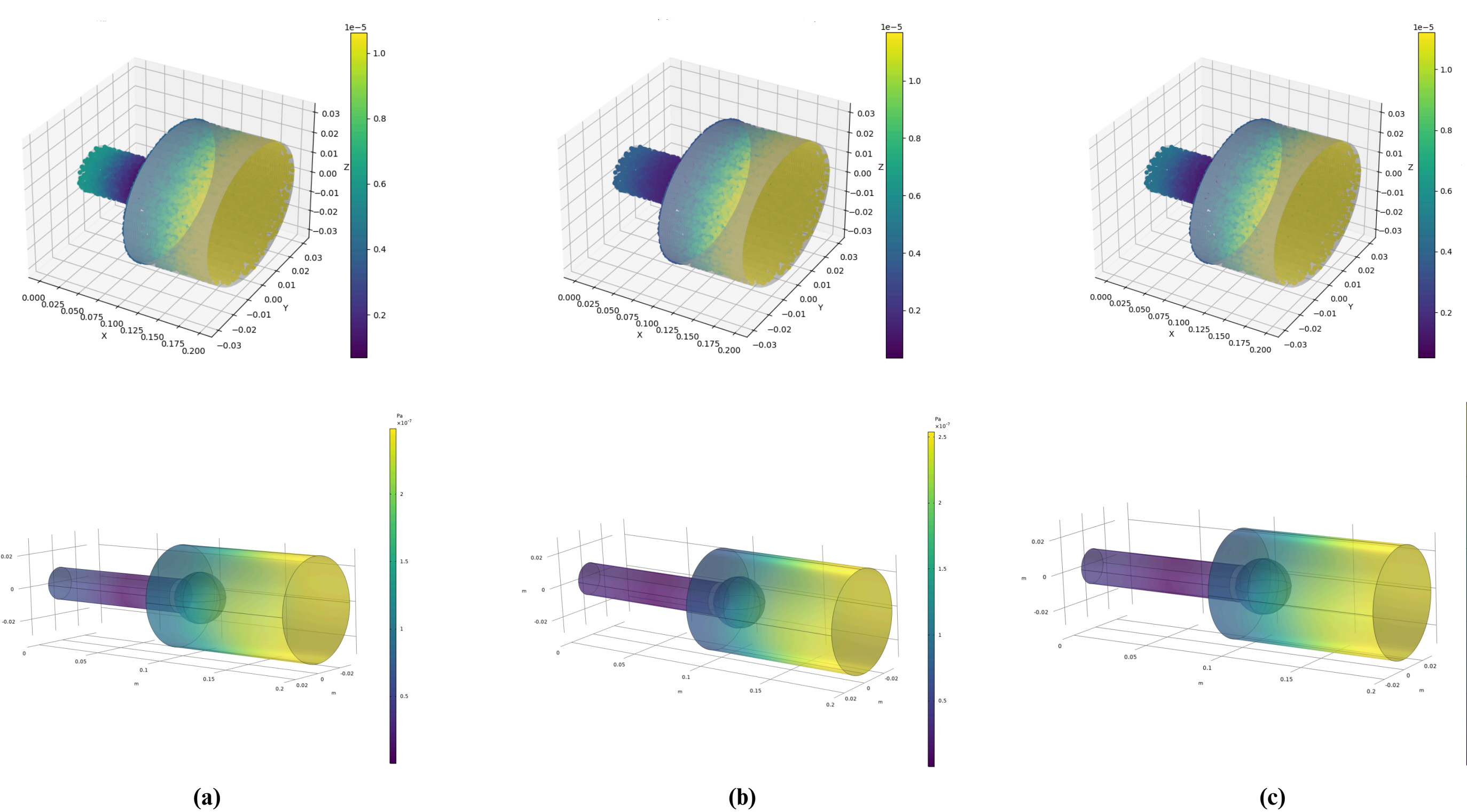


Figure 5: Comparison of pressure field obtained from the Helmholtz solver (bottom) and Green's function formulation (top) for (a) Hydrogen, (b) Ammonia and (c) 1:1 (by moles) mixture of Hydrogen and Ammonia as fuels and air as oxidizer having equivalence ratio 0.6 at 2500 Hz

4. Conclusions

Simulations were performed using the Helmholtz solver in order to determine the eigenfunctions of a combustor model with a double duct geometry. The eigenfrequencies thus obtained were validated against the analytical solution found using the transfer matrix method. Based on the eigenmode analysis, the Green's function was formulated for the model combustor, using which the pressure field was determined and compared against the pressure field obtained from the Helmholtz equation solver. When the domain material is taken to be air, the pressure fields obtained by both the methods are similar in the order of magnitude and trend as shown in figure 3. Eigenmode analysis of the model with unreacted fuel-air mixture at the injector and reacted gases in the combustor shows insignificant variation of the eigenfunctions with equivalence ratio (figure 1). Analysis of the obtained pressure fields is thus performed for an equivalence ratio of 0.6, with Hydrogen, Ammonia and 1:1 molar mixture of Hydrogen and Ammonia considered as fuels as shown in figures 4 & 5. The pressure fields obtained from the Helmholtz solver show highly similar trends to those obtained from the Green's function based reduced order model.

References

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